Multi-Cloud Disaster Recovery System

Project Phase Plan

4-Phase Implementation Roadmap

Document Type: Phase Breakdown

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Industry Skills: AWS - Azure - Disaster Recovery - Automation - High Availability

Overview

The project is broken into four sequential phases, moving from foundational multi-region setup through data replication, automated failover, and finally monitoring/orchestration polish plus testing. Each phase builds on the previous one and ends with a demonstrable milestone.

Phase	Focus	Approx. Duration
Phase 1	Foundation - Multi-Region Deployment & Backups	Week 1-2
Phase 2	Data Layer - Database Replication & Storage Sync	Week 3-4
Phase 3	Automation - Health Monitoring & Failover Orchestration	Week 5-6
Phase 4	Resilience - Recovery Orchestration, Failback & Testing	Week 7-8

Phase 1: Foundation - Multi-Region Deployment & Backups

Duration: Week 1-2

Objective: Stand up identical application infrastructure in a primary AWS region and a secondary region (Azure or a second AWS region), and establish automated backup routines.

Key Tasks

- Design the base architecture and choose primary/secondary regions and providers.
- Write Infrastructure-as-Code templates (Terraform) for compute, networking, and storage.
- Deploy the application stack identically in both regions using CI/CD pipelines.
- Configure automated, scheduled backups for databases and object storage.
- Set up IAM/RBAC roles and basic security groups/NSGs in both regions.

Deliverables

- Working multi-region infrastructure (IaC scripts checked into version control).
- Automated backup schedule with defined retention policy.
- Architecture diagram of the base (pre-replication) deployment.

Industry Skills Practiced

AWS fundamentals, Azure fundamentals, Infrastructure as Code, High Availability basics.

Phase 2: Data Layer - Database Replication & Storage Sync

Duration: Week 3-4

Objective: Implement continuous database replication and storage synchronization between the primary and secondary regions to minimize RPO.

Key Tasks

- Configure cross-region database replication (e.g., Aurora Global Database, Azure SQL auto-failover groups, or logical replication/CDC).
- Set up cross-region/cross-cloud object storage replication (S3 CRR, Azure Blob GRS, or a sync job).
- Instrument and export replication lag as a monitored metric.
- Test manual promotion of the standby database to confirm replication integrity.
- Document RPO measurements from initial replication tests.

Deliverables

- Live database replication with measured lag under target RPO.
- Replicated object storage between regions.
- Replication lag dashboard/metric feed.

Industry Skills Practiced

Database Replication, Disaster Recovery fundamentals, Data consistency patterns.

Phase 3: Automation - Health Monitoring & Failover Orchestration

Duration: Week 5-6

Objective: Build the health monitoring layer and the automated failover orchestration engine that detects primary region failure and redirects traffic without manual intervention.

Key Tasks

- Implement multi-signal health checks (application, database, compute, network).
- Build the composite health scoring and confirmation-window logic.
- Build the failover orchestration state machine (Step Functions / Logic Apps).
- Integrate DNS/traffic manager failover routing (Route 53 / Azure Traffic Manager).
- Implement automated notifications for health state changes and failover events.
- Run first end-to-end simulated failover drill.

Deliverables

- Working automated failover triggered by a simulated primary-region failure.
- Health monitoring dashboard showing live region status.
- Notification integration (email/Slack) firing on DR events.
- First measured RTO from a simulated drill.

Industry Skills Practiced

Automation, Health Monitoring, Failover Automation, AWS/Azure orchestration services.

Phase 4: Resilience - Recovery Orchestration, Failback & Testing

Duration: Week 7-8

Objective: Complete the recovery lifecycle with controlled failback, add resilience polish (manual override, audit logging), and run comprehensive DR testing.

Key Tasks

- Implement failback orchestration: re-sync primary region and shift traffic back safely.
- Add manual override controls (trigger/abort failover) with role-based access.

- Build the full DR event audit log and history view in the dashboard.
- Implement automated backup restore-drill testing.
- Run multiple full-cycle DR simulations to validate RTO/RPO against targets.
- Write operational runbooks and final project documentation/demo.

Deliverables

- End-to-end recovery orchestration including tested failback.
- Audit log and dashboard reporting DR readiness and event history.
- Final validated RTO/RPO metrics from repeated test drills.
- Runbook documentation and final project demo/presentation.

Industry Skills Practiced

Recovery Orchestration, Disaster Recovery testing, High Availability validation, Operational documentation.

Phase Dependency Summary

Phase	Depends On	Unlocks
Phase 1 - Foundation	None	Base infrastructure required by all later phases
Phase 2 - Data Layer	Phase 1 complete	Replicated data needed for real failover testing
Phase 3 - Automation	Phase 1 & 2 complete	Automated detection and traffic redirection
Phase 4 - Resilience	Phase 1, 2 & 3 complete	Full DR lifecycle: failover + failback + validation